

# DS701 Homework 1: Question Types; Vectors, Norms, Dot Products

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## SOLUTIONS

### Problem 1: Classify the question

*Practices: naming the question type before choosing a technique (Lecture 01).*

A music-streaming service is planning its analytics for the fall. Six questions come up in the planning meeting:

1. What was the median listening time per user last month, broken down by country?
  2. Do our users fall into a few natural groups by the genres they listen to — something worth a closer look?
  3. Across our whole user base (not just the 2,000 users we surveyed), do people who listen during a commute play more podcasts than people who do not?
  4. Will *this* user cancel their subscription in the next 30 days?
  5. If we give every free-trial user a personalized playlist on day 1, will more of them convert to paid?
  6. By exactly which route does a personalized playlist raise conversion — does it lengthen sessions, which builds a daily habit, which then drives payment?
- (a) Classify each of the six questions using the lecture’s taxonomy (descriptive, exploratory, inferential, predictive, causal, mechanistic). For each, give **one line** saying what data and what kind of analysis would answer it (a summary table, an experiment, a model scored on held-out users, ...). Name the word or phrase in the question that decided your classification.
- (b) Question 4 is predictive. Rewrite it as a **causal** question about the same subscription decision — one sentence. Then say, in two or three sentences, what changes: what counts as evidence for the predictive version, what counts as evidence for the causal version, and why a very accurate churn model does not by itself answer the causal one.
- (c) A colleague proposes answering question 5 by comparing conversion rates of past trial users who *happened* to build their own playlist on day 1 against those who did not, and reports “playlist users converted at twice the rate”. Which question type does this comparison actually answer? Name one plausible reason the two groups differ that has nothing to do with the playlist, and say what design would answer question 5 as asked. Keep it to a short paragraph.

**Solution**

- (a) One defensible key (the deciding phrase in *italics*; other readings earn credit if argued):
1. **Descriptive** — *median ... last month, by country*: a summary of data we already have; a group-by table. No claim beyond the users in the table.
  2. **Exploratory** — *natural groups ... worth a closer look*: looking for structure with no labels; a clustering (k-means/hierarchical) on genre-listening vectors, followed by plots. Generates hypotheses, does not test them.
  3. **Inferential** — *across our whole user base (not just the ... surveyed)*: does a pattern in a sample hold in the population? Compare the two groups' podcast rates in the survey with a confidence interval or test for the difference.
  4. **Predictive** — *will this user ... next 30 days*: a forecast for a new case; a classifier trained on past users' features and churn outcomes, judged by accuracy on held-out users.
  5. **Causal** — *if we give ... will more of them convert*: what happens under an intervention; a randomized experiment (A/B test) that assigns the playlist to a random half of trial users.
  6. **Mechanistic** — *by exactly which route*: how the effect operates; measure the intermediate steps (session length, daily use) and test whether the effect flows through them — the most demanding rung.
- (b) Causal rewrite, e.g. “If we offer *this* user a discount (or a reminder, or a new feature) this month, does that reduce the chance they cancel?” — the key change is an intervention we control. Evidence for the predictive version: held-out accuracy — the model is right about who cancels among users it has not seen. Evidence for the causal version: that the comparison is free of confounding — ideally an experiment where the intervention is randomly assigned. An accurate churn model tells you *which* users are at risk, not *what would happen if you acted*: the features it uses (say, “has not opened the app in 10 days”) are correlated with cancelling but forcing them to change need not change the outcome. This is the lecture’s “type error”.
- (c) The comparison answers an **inferential/descriptive** association question — “do users who built playlists convert more?” — not the causal one asked. Users who build a playlist on day 1 are already the engaged ones (more time, more interest in music, more likely to pay anyway); that engagement, not the playlist, can explain the doubled rate. To answer question 5, randomize: give the playlist to a random half of new trial users and withhold it from the other half, then compare conversion. Randomization makes the two groups alike in everything except the playlist.

*Common mistake*: calling question 3 descriptive because it “just compares two averages” — the phrase “across our whole user base” makes it a claim beyond the sample.

**Problem 2: Two vectors by hand**

*Practices: norms, dot product, cosine similarity, unit vectors (A7).*

Let  $\mathbf{x} = (1, 2, 2)$  and  $\mathbf{y} = (2, 2, 1)$  in  $\mathbb{R}^3$ . Definitions you need:  $\|\mathbf{v}\|_1 = \sum_i |v_i|$ ,  $\|\mathbf{v}\|_2 = \sqrt{\sum_i v_i^2}$ ,  $\|\mathbf{v}\|_\infty = \max_i |v_i|$ ,  $\mathbf{x} \cdot \mathbf{y} = \mathbf{x}^T \mathbf{y} = \sum_i x_i y_i$ , and  $\cos(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x}^T \mathbf{y}}{\|\mathbf{x}\|_2 \|\mathbf{y}\|_2}$ . A *unit vector* has  $\ell_2$  norm 1.

- (a) Compute  $\|\mathbf{x}\|_1$ ,  $\|\mathbf{x}\|_2$ ,  $\|\mathbf{x}\|_\infty$  and the same three norms of  $\mathbf{y}$ .

- (b) Compute  $\mathbf{x} \cdot \mathbf{y}$ , the cosine similarity  $\cos(\mathbf{x}, \mathbf{y})$  (as an exact fraction), and the angle between them (leave it as arccos of your fraction, and give degrees if you have a calculator). Then write down the unit vector in the direction of  $\mathbf{x}$  and verify by direct computation that its  $\ell_2$  norm is 1.

- (c) Show that for **every**  $\mathbf{v} \in \mathbb{R}^n$ ,

$$\|\mathbf{v}\|_\infty \leq \|\mathbf{v}\|_2 \leq \|\mathbf{v}\|_1.$$

A short argument is expected (three or four lines): for the left inequality, compare  $\max_i v_i^2$  with  $\sum_i v_i^2$ ; for the right, expand  $(\sum_i |v_i|)^2$ . Check the chain on your numbers from (a).

- (d) Give a nonzero vector for which all three norms are **equal**, and say in one line what has to be true of a vector for that to happen.

### Solution

- (a)  $\|\mathbf{x}\|_1 = 1 + 2 + 2 = 5$ ;  $\|\mathbf{x}\|_2 = \sqrt{1 + 4 + 4} = \sqrt{9} = 3$ ;  $\|\mathbf{x}\|_\infty = 2$ . By symmetry  $\|\mathbf{y}\|_1 = 5$ ,  $\|\mathbf{y}\|_2 = \sqrt{4 + 4 + 1} = 3$ ,  $\|\mathbf{y}\|_\infty = 2$ .
- (b)  $\mathbf{x} \cdot \mathbf{y} = 1 \cdot 2 + 2 \cdot 2 + 2 \cdot 1 = 8$ .  $\cos(\mathbf{x}, \mathbf{y}) = \frac{8}{3 \cdot 3} = \frac{8}{9} \approx 0.889$ ; angle  $\theta = \arccos(8/9) \approx 0.476$  rad  $\approx 27.3^\circ$  — pointing in nearly the same direction. Unit vector:  $\mathbf{x}/\|\mathbf{x}\|_2 = (1/3, 2/3, 2/3)$ ; its norm is  $\sqrt{1/9 + 4/9 + 4/9} = \sqrt{9/9} = 1$ . (In general  $\|\mathbf{x}/c\|_2 = \|\mathbf{x}\|_2/c$  for  $c > 0$  — the norm's scaling property.)
- (c) Let  $M = \max_i |v_i| = \|\mathbf{v}\|_\infty$ . *Left:*  $\|\mathbf{v}\|_2^2 = \sum_i v_i^2 \geq M^2$  because the sum of nonnegative terms is at least its largest term; take square roots. *Right:*  $\|\mathbf{v}\|_1^2 = (\sum_i |v_i|)^2 = \sum_i v_i^2 + \sum_{i \neq j} |v_i||v_j| \geq \sum_i v_i^2 = \|\mathbf{v}\|_2^2$ , since the cross terms are nonnegative; take square roots. Check on  $\mathbf{x}$ :  $2 \leq 3 \leq 5$ . ✓
- (d) Any vector with exactly one nonzero entry, e.g.  $\mathbf{v} = (0, -5, 0)$ : all three norms equal 5. That is exactly when both inequalities in (c) are equalities: one entry carries the whole sum (so  $\sum v_i^2 = M^2$ ) and every cross term  $|v_i||v_j|$ ,  $i \neq j$ , is zero.

*Common mistake:* writing the cosine as  $8/\sqrt{9}$  — the denominator is the product of the two norms,  $3 \cdot 3 = 9$ , not a single norm.

### Problem 3: Data as a matrix

*Practices:* rows = objects, columns = features; centering; the dot product of two feature columns as a first measure of “moving together”.

Four students report hours studied and their exam score:

| student | hours ( $h$ ) | score ( $s$ ) |
|---------|---------------|---------------|
| 1       | 2             | 60            |
| 2       | 4             | 70            |
| 3       | 6             | 70            |
| 4       | 8             | 80            |

Written as a data matrix  $X$  this is  $4 \times 2$ :  $m = 4$  objects (rows),  $n = 2$  features (columns). To **mean-center** a column, subtract that column's mean from each of its entries.

- (a) Compute the mean of each column and write out the centered matrix  $X_c$  (the two centered columns  $\mathbf{h}_c$  and  $\mathbf{s}_c$ ). Verify that each centered column sums to zero, and explain in one line why that must always happen.
- (b) Compute the dot product  $\mathbf{h}_c \cdot \mathbf{s}_c$  of the two centered columns. What does its **sign** tell you about how the two features move together across students? (Think about which students contribute positive terms and which negative.)
- (c) Compute the cosine similarity of  $\mathbf{h}_c$  and  $\mathbf{s}_c$  (three decimals;  $\sqrt{4000} \approx 63.25$ ). What would a value near 1, near 0, or near  $-1$  say about hours and scores?
- (d) Suppose the scores had been recorded as fractions (0.60, 0.70, 0.70, 0.80) instead of percentages. Without redoing everything, say what happens to the dot product in (b) and to the cosine in (c), and one sentence on why that makes the cosine the more useful of the two as a summary of “how related”.

### Solution

(a)  $\bar{h} = (2 + 4 + 6 + 8)/4 = 5$ ,  $\bar{s} = (60 + 70 + 70 + 80)/4 = 70$ .

$$\mathbf{h}_c = (-3, -1, 1, 3), \quad \mathbf{s}_c = (-10, 0, 0, 10), \quad X_c = \begin{bmatrix} -3 & -10 \\ -1 & 0 \\ 1 & 0 \\ 3 & 10 \end{bmatrix}.$$

Both columns sum to 0:  $\sum_i (v_i - \bar{v}) = \sum_i v_i - m\bar{v} = m\bar{v} - m\bar{v} = 0$  by the definition of the mean.

- (b)  $\mathbf{h}_c \cdot \mathbf{s}_c = (-3)(-10) + (-1)(0) + (1)(0) + (3)(10) = 30 + 0 + 0 + 30 = 60$ . Positive: students below the mean in hours (1) are also below in score, and above (4) is above — the two features move in the same direction. A negative product would mean opposite directions; near zero, no consistent linear pattern.
- (c)  $\|\mathbf{h}_c\|_2 = \frac{\sqrt{9+1+1+9}}{60} = \frac{\sqrt{20}}{60}$ ,  $\|\mathbf{s}_c\|_2 = \frac{\sqrt{100+0+0+100}}{60} = \frac{\sqrt{200}}{60}$ , so  $\cos = \frac{\sqrt{20}\sqrt{200}}{\sqrt{4000}} = \frac{60}{63.25} \approx 0.949$ . Near 1: strong same-direction (nearly linear increasing) relationship; near 0: no linear relationship; near  $-1$ : strong opposite-direction relationship. (This number *is* the Pearson correlation of hours and score — you will meet it under that name in Lecture 05.)
- (d) The centered score column becomes  $\mathbf{s}_c/100$ , so the dot product shrinks by the same factor to  $60/100 = 0.6$ , while the cosine is unchanged at 0.949: both the numerator and  $\|\mathbf{s}_c\|$  scale by  $1/100$  and the factor cancels. The dot product’s magnitude depends on the units the features happen to be recorded in; the cosine does not — it measures direction only, so it is a unit-free summary of how related the two features are.

*Common mistake:* mean-centering the rows (students) instead of the columns (features). Centering is per feature.